

CENG381 Stochastic Processes

Irreducibility and Periodicity — Practice 1 (Answer Key)

Q1 Solution

a) Transition diagram and irreducibility:

The diagram is a single directed 4-cycle: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$.

Irreducible: from state i , after $j-i \pmod{4}$ steps you reach state j . Every pair of states communicates. ✓

b) Period of state 1:

$P^t(1,1) > 0$ iff t is a multiple of 4 (the chain returns to 1 only after completing full 4-cycles).

$T(1) = \{4, 8, 12, 16, \dots\}$

$$\text{period}(1) = \gcd\{4, 8, 12, \dots\} = 4$$

c) P^4 and P^5 :

Since the chain traverses the 4-cycle deterministically, after exactly 4 steps every state returns to itself:

$$P^4 = I \quad (4 \times 4 \text{ identity matrix})$$

$$P^5 = P \quad (\text{one full cycle plus one extra step})$$

The chain does NOT converge to $\pi = (1/4, 1/4, 1/4, 1/4)$. Starting from state 1, the distribution at time t is always a point mass: it cycles $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1 \rightarrow \dots$ deterministically. A periodic chain of period $d > 1$ oscillates; the limit $\lim_{n \rightarrow \infty} P^n$ does not exist (it depends on $n \pmod{4}$).

Q2 Solution

a) Transition matrix and irreducibility:

	0	1	2	3
0	[0	1	0	0]
1	[0	0	1	0]
2	[0	0	0	1]
3	[1/2	0	1/2	0]

Irreducible: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$ (prob $1/2$) and $3 \rightarrow 2 \rightarrow 3$ (prob $1/2$). Every state can reach state 0 via the path $\dots \rightarrow 3 \rightarrow 0$. State 0 can reach any state by following $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow \dots \rightarrow$. All states communicate. ✓

b) Period of state 0:

A path must start and end at 0. Possible return paths:

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$ (length 4, prob $1/2$)

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$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 0$ (length 6, prob $1/4$)

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 0$ (length 8, prob $1/8$), ...

So $T(0) = \{4, 6, 8, 10, \dots\}$ (all even integers ≥ 4).

$$\text{period}(0) = \gcd\{4, 6, 8, 10, \dots\} = \gcd(4, 6) = 2$$

The chain has period 2. (Even steps stay in $\{0, 2\}$; odd steps stay in $\{1, 3\}$.)

c) Stationary distribution:

Balance equations $\pi \cdot P = \pi$:

$$\pi_0 = (1/2) \cdot \pi_3 \quad \dots \text{ (I)}$$

$$\pi_1 = \pi_0 \quad \dots \text{ (II)}$$

$$\pi_2 = \pi_1 + (1/2) \cdot \pi_3 \quad \dots \text{ (III)}$$

$$\pi_3 = \pi_2 \quad \dots \text{ (IV)}$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \quad \dots \text{ (V)}$$

From (I): $\pi_3 = 2\pi_0$

From (II): $\pi_1 = \pi_0$

From (IV): $\pi_3 = \pi_2 \rightarrow \pi_2 = 2\pi_0$

From (V): $\pi_0 + \pi_0 + 2\pi_0 + 2\pi_0 = 1 \rightarrow 6\pi_0 = 1$

$$\pi = (1/6, 1/6, 1/3, 1/3)$$

$$E[\text{return time to } 0] = 1/\pi_0 = 6 \text{ steps}$$

Q3 Solution

a) Lazy chain $Q = (1/2) \cdot I + (1/2) \cdot P$:

	0	1	2	3
0	[1/2	1/2	0	0]
1	[0	1/2	1/2	0]
2	[0	0	1/2	1/2]
3	[1/4	0	1/4	1/2]

b) Aperiodicity of state 0 in Q:

$$Q(0,0) = (1/2) \cdot I(0=0) + (1/2) \cdot P(0,0) = 1/2 + 0 = 1/2 > 0$$

Since $Q(0,0) > 0$, $T(0)$ contains $t=1$.

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$\gcd(T(0))$ divides 1 $\rightarrow$ period(0) in $Q = 1$ (aperiodic). ✓

c) Stationary distribution of Q :

Suppose $\pi \cdot P = \pi$. Then:

$$\begin{aligned}\pi \cdot Q &= \pi \cdot [(1/2) \cdot I + (1/2) \cdot P] \\ &= (1/2) \cdot (\pi \cdot I) + (1/2) \cdot (\pi \cdot P) \\ &= (1/2) \cdot \pi + (1/2) \cdot \pi \\ &= \pi \quad \checkmark\end{aligned}$$

So $\pi = (1/6, 1/6, 1/3, 1/3)$ is also the stationary distribution of Q .

Long-run fractions: state 0 $\rightarrow$ 16.7%, state 1 $\rightarrow$ 16.7%, state 2 $\rightarrow$ 33.3%, state 3 $\rightarrow$ 33.3%.

Unlike the original chain, Q is aperiodic and irreducible, so $P^n(Q)$ converges to π regardless of starting state.